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YEAR: Third Year

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PROJECT NAME: LAPTOP PRICE PREDICTOR

A comprehensive dataset detailing various laptop features like brand, CPU, RAM, storage, and screen size was used. This data underwent meticulous preparation for analysis. Missing values were addressed, categorical features encoded numerically, and the entire dataset scaled to ensure consistency. This meticulous data preparation ensures the models can effectively learn the underlying relationships between features and prices.

With the prepared data, the project evaluated various machine learning models for their accuracy in price prediction. This included established methods like linear regression alongside more complex algorithms like support vector machines and extra trees regression. Each model underwent rigorous training on a portion of the data using cross-validation to prevent overfitting. The evaluation process relied on metrics like mean squared error, which measures the average squared difference between predicted and actual prices. This suggests machine learning's effectiveness in capturing the relationship between laptop specifications and market value, potentially benefiting both consumers and retailers.

Features

- Preprocessing Data
- EDA
- Feature Engineering
- Machine Learning
- Prediction
- Hosted on Streamlit

MODEL DEMO:

The image shows a laptop configuration tool on the left and an Amazon product page on the right. The configuration tool has a dark theme and lists various components with dropdown menus. The predicted price for the selected configuration is 56742. The Amazon page shows a Lenovo IdeaPad Gaming laptop with specifications: 3 AMD Ryzen 5 5600H, 15.6" (39.62cm) FHD IPS 120Hz Gaming Laptop, 8GB/256GB SSD+1TB/, Windows 11 Home /NVIDIA GTX 1650 4GB/Alexa/3 Month Game Pass/Shadow Black/2.2Kg, 82K201Y9IN. The price is \$52,700 with a 41% discount. The configuration tool on the left has the following settings:

- CPU: AMD Processor
- HDD(In GB): 256
- SSD(In GB): 1024
- GPU: Nvidia
- OS: Windows

The predicted price of this configuration is 56742.

The Amazon page shows the following details for the Lenovo IdeaPad Gaming laptop:

- Lenovo IdeaPad Gaming 3 AMD Ryzen 5 5600H 15.6" (39.62cm) FHD IPS 120Hz Gaming Laptop 8GB/256GB SSD+1TB/ Windows 11 Home /NVIDIA GTX 1650 4GB/Alexa/3 Month Game Pass/Shadow Black/2.2Kg, 82K201Y9IN
- Visit the Lenovo Store
- 4.5 stars (143 ratings)
- 100 answered questions
- 41% off \$52,700
- Incl. of all taxes. EMI starts at ₹2,518. No Cost EMI available. EMI options: --

The image shows a laptop configuration tool on the left and an Apple product page on the right. The configuration tool has a dark theme and lists various components with dropdown menus. The predicted price for the selected configuration is 109170. The Apple page shows a 2022 MacBook Air Laptop with M2 chip: 34.46 cm (13.6-inch) Liquid Retina Display, 8GB RAM, 256GB SSD Storage, Backlit Keyboard, 1080p FaceTime HD Camera. Works with iPhone/iPad; Space Grey. The price is \$1,13,990 with a 5% discount. The configuration tool on the left has the following settings:

- Intel Core i7
- HDD(In GB): 0
- SSD(In GB): 256
- GPU: Intel
- OS: Mac

The predicted price of this configuration is 109170.

The Apple page shows the following details for the 2022 MacBook Air:

- Apple 2022 MacBook Air Laptop with M2 chip: 34.46 cm (13.6-inch) Liquid Retina Display, 8GB RAM, 256GB SSD Storage, Backlit Keyboard, 1080p FaceTime HD Camera. Works with iPhone/iPad; Space Grey
- Visit the Apple Store
- 4.5 stars (18 ratings)
- 6 answered questions
- 5% off \$1,13,990

Github Link to the project: <https://github.com/tanishagandhi/Laptop-Price-Predictor>